Introduction to Apache Web Server

Session 1



For Aptech Center Use Only

© Aptech Limited.

Objectives

- *Describe Apache Web Server*
- *Discuss the history of Apache Web Server*
- *Explain the features of Apache Web Server*
- *Explain the Architecture of Apache Web Server*

For Aptech Center Use Only

© Aptech Limited.

Apache Web Server

- Is a software application that uses `HTTP` protocol over the World Wide Web
- Deliver the required resources to the user
- Is the most commonly used Web server on the Internet
- Is `HTTP/1.1` compliant

For Aptech Center Use Only

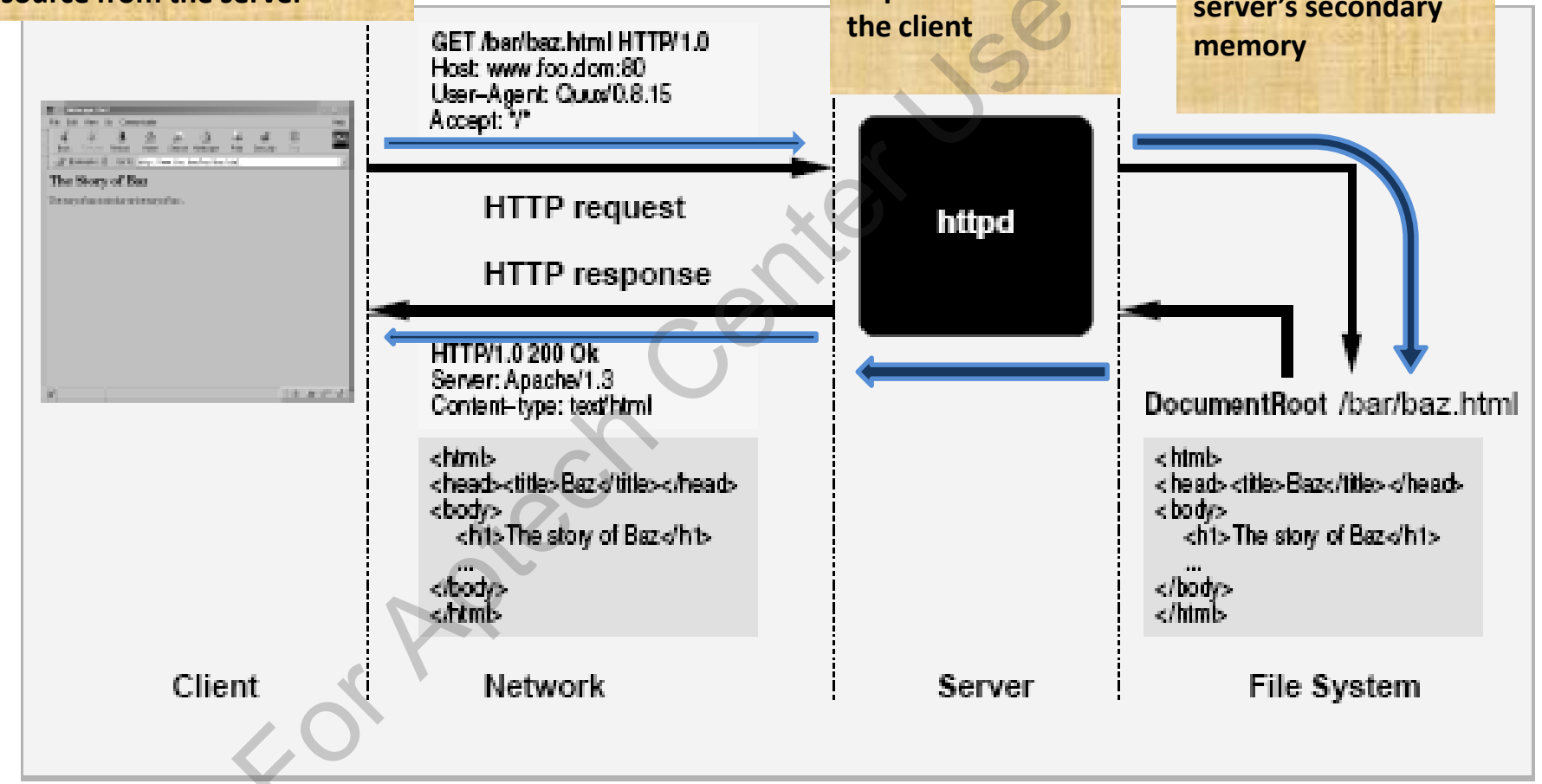
© Aptech Limited.

Apache Web Server

Using HTTP, the client initiates communication by requesting a resource from the server

The server processes the resource request and sends the requested file to the client

The resource is a file located in the server's secondary memory



For Aptech Center Use Only

© Aptech Limited.

Different Types of Web Servers

- Apache HTTP Server
- Internet Information Server (IIS)
- Sun ONE
- National Center for Supercomputing Applications (NCSA) Server
- America Online's (AOL) Server

For Aptech Center Use Only

© Aptech Limited.

Apache HTTP Server

- Is a modular software as it allow users to install only those modules that are required for their Websites
- Language modules supported – `Perl`, `Python`, `Tcl`, and `PHP` scripts
- Authentication modules supported – `.htaccess`, `mod_access`, `mod_auth`, `mod_digest`, and `mod_auth_digest`
- URL rewriter – `mod_rewrite`
- Allows accessing the log files through a Web browser using free scripts, such as `AWStats`

For Aptech Center Use Only

© Aptech Limited.

History of Apache Web Server

- Developed at the National Center for Supercomputing Application (NCSA)
- On December 1st 1995, was released
- Version 2.2 was introduced with:
  - New structure for authentication
  - Improved caching modules
  - Support for proxy load balancing
  - Improved Web server capacity

For Aptech Center Use Only

© Aptech Limited.

Features of Aptech Web Server

- Support for the latest `HTTP 1.1` protocol
- Authentication/Authorization Modules
- Configuration Files
- Graceful stop
- Proxying
- Smart Filtering
- Large File Support
- SQL Database support
- File-based configuration

For Aptech Center Use Only

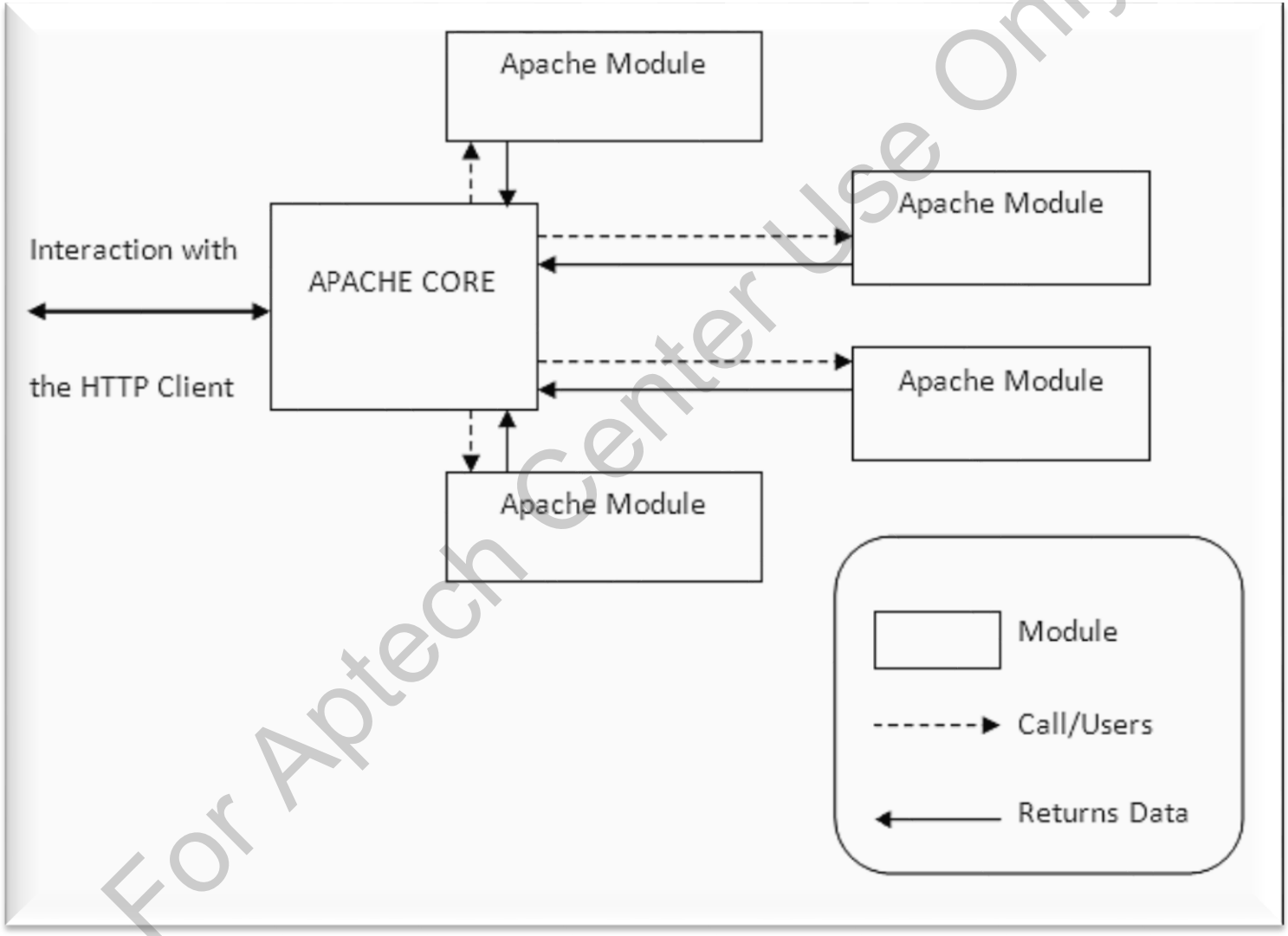
© Aptech Limited.

- Support for Common Gatewaay Interface (CGI)
- Support for Fast CGI
- Support for virtual hosts
- Support for HTTP authentication
- Integrated Perl
- Support for PHP scripting
- Java Servlet support
- Server status and customizable logs
- Support for Server-Side Includes (SSI) and Secure Socket Layer (SSL)

For Aptech Center Use Only

© Aptech Limited.

Architecture of Apache Web Server



For Aptech Center Use Only

© Aptech Limited.

- The core contains the basic functionality of the server
- The different components of the core are as follows:
  - `http_protocol.c` - contains functions that communicate with the client and transfers data to the client using this component
  - `http_main.c` - contains the functionality for starting up the server and waits for and accepts connections

For Aptech Center Use Only

© Aptech Limited.

- `http_request.c` - contains the logic for handling request processing, dispatching control to the modules, and for error handling
- `http_core.c` - contains the logic for implementing the basic functionality
- `alloc.c` - contains the functionality for allocating resource pools and keeping track of resources
- `http_config.c` - contains the logic for reading and managing the information present in the configuration files

For Aptech Center Use Only

© Aptech Limited.

- Multi-Processing Modules (MPM):
  - Used to isolate tasks in a program
  - Parent process was split into a set of child processes that served the actual request
  - Parent processes:
    - Monitors the child processes
    - Spawns or kills child processes depending upon the request received
  - Server process and services request are also started with MPM

For Aptech Center Use Only

© Aptech Limited.

- Some of the MPMs are as follows:
  - Prefork
    - Creates a pool of child processes to handle the request
  - Worker
    - Enables thread support
  - Filtering I/O
    - Features layered I/O
    - CGI script produces output as SSL tags which are then processed before final output is sent to the browser
  - CGI Daemon
    - Runs in background and defines how Web servers delegate the generation of Web pages to CGI scripts

For Aptech Center Use Only

© Aptech Limited.

- CGI Daemon
  - Required to execute `Perl` and `PHP` scripts on the Web browser
  - Uses the `mod_cgid` module to interact with threads

For Aptech Center Use Only

© Aptech Limited.

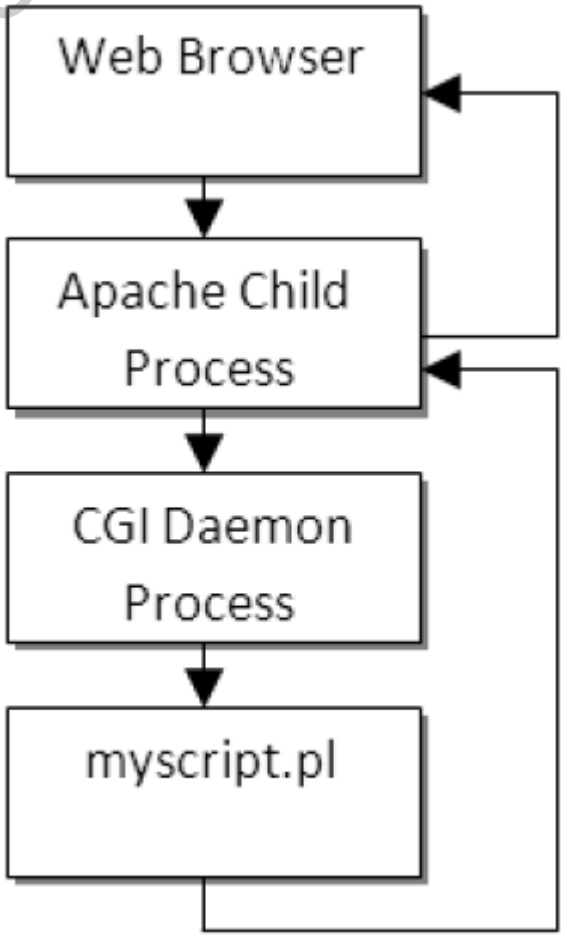
- Working of CGI Daemon

The Web browser creates the child process

When a CGI reads a thread within a child process, it passes the request to the CGI daemon

The CGI daemon spawns the CGI script and the CGI script generates data to the thread in the child process

The thread returns the data to the Web browser



For Aptech Center Use Only

© Aptech Limited.

Summary

- A Web server is a software application that uses the *HTTP*
- Apache contains a single primary configuration file called *httpd.conf*
- Each MPM starts with the server process and services requests through child processes or threads depending upon which MPM is implemented
- The `prefork` *MPM* creates a pool of *child processes* to handle the requests
- The *threaded MPM process* enables a predefined number of threads
- Each thread within a child process serves a different request

For Aptech Center Use Only

© Aptech Limited.